THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A stone is dropped into a well and the splash is heard 2.1 s later. If the speed of sound is 340 m/s and g = 10, the depth is closest to: 1

(a) 25 m **(b)** 15 m **(c)** 35 m **(d)** 20 m

2. A listener hears a hammer blow on a long steel rail twice: first through the rail and then through air. The sound through the rail comes first because: 1

(a) air absorbs sound **(b)** the rail is shorter

(c) the speed of sound in steel is greater **(d)** the speed in air is greater

3. Assertion (A): The pitch of a sound increases with its frequency. Reason (R): A high frequency means more vibrations per second. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

4. A stone is dropped into a well 45 m deep (g = 10) and the splash is heard after (v = 340 m/s) about: 1

(a) 3 s **(b)** 3.3 s **(c)** 3.13 s **(d)** 0.13 s

5. Two sound waves have wavelengths 0.5 m and 1.0 m in air. The ratio of their frequencies is: 1

(a) 1 : 2 **(b)** 4 : 1 **(c)** 1 : 1 **(d)** 2 : 1

6. A sound wave takes 1 s in air and 0.23 s in steel over the same distance. The speed in steel is about (v air = 340 m/s): 1

(a) 78 m/s **(b)** 340 m/s **(c)** 1478 m/s **(d)** 4000 m/s

7. Assertion (A): In a big empty hall the sound is not clear. Reason (R): Repeated reflections produce reverberation. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. A source of frequency 100 Hz sends waves at 300 m/s. The distance between a compression and the next compression is: 1

(a) 0.33 m **(b)** 1.5 m **(c)** 30 m **(d)** 3 m

9. A sound wave produced by a source has frequency 10 Hz in air. A man will: 1

(a) hear it as a low note **(b)** hear it as ultrasound

(c) hear it as a high note **(d)** not hear it as it is infrasonic

10. An echo returns in 0.2 s at 20 degrees Celsius, and the same surface gives an echo at 0.19 s at 34 degrees Celsius. Which is true? 1

(a) Distance changed **(b)** Speed is unchanged

(c) Speed is lower at 34 degrees Celsius **(d)** Speed is higher at 34 degrees Celsius

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A sonar pulse reflected from a submarine returns after 0.4 s. Find the distance of the submarine (v = 1500 m/s). 2

12. An observer measures thunder 8 s after lightning. Estimate the distance and explain why the light time is negligible. 2

13. Why does the pitch of a person's voice change when he speaks after inhaling helium? Explain with v = f lambda. 2

14. Why do we not hear an echo of our voice from the ceiling of a small room? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The pulse of a sonar returns from the sea bed after 3.6 s in sea water (v = 1540 m/s). Find the depth. If the pulse is echoed also from a shoal at 0.9 s, find the height of the shoal above the bed. 3

16. Calculate the number of vibrations per minute of a source emitting an audible sound of lowest and highest frequencies and the wavelengths in air (v = 340 m/s). 3

17. A pulse of ultrasound is sent into the body and reflections are received at 10 and 12 microseconds after emission. If v = 1540 m/s, find the depths of the two reflecting layers and the thickness between them. 3

OR

A sound of frequency 256 Hz is produced in air at 20 degrees Celsius. Find the wavelength. If the temperature changes so that the speed becomes 350 m/s find the new wavelength (v1 = 344 m/s). 3

18. Why does the sound of a railway train appear louder at night than during the day? 3

OR

Why do we hear our own voice differently in a recording than when we speak? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A ship's sonar receives an echo from an iceberg after 6 s and an echo from the sea bed after 8 s in water of v = 1450 m/s. Find the distance of the iceberg, the depth of the sea bed and the height of the iceberg top above the bed if they are on the same line below the ship. 5

OR

The frequency of a siren is 600 Hz. Find its time period and wavelength in air (340 m/s). If the same siren is sounded under water (1500 m/s) find the wavelength and the number of waves in 1 s in both media. 5

20. Explain loudness, pitch and quality of musical sounds. Compare a flute and a drum on frequency range, amplitude and sustained quality, and the effect of striking the drum harder. 5

OR

A person standing between two walls claps and hears two echoes, the first after 0.6 s and the second after 1.0 s. Find the distances of the walls (v = 340 m/s), the distance between the walls and the time at which the person would hear a single combined echo if he moved to the middle. 5

END OF QUESTION PAPER — Total: 40 Marks